

GaitDynamics: a generative foundation model for analyzing human walking and running

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Understanding the dynamics of human gait, including both motions and forces, is vital to promote human mobility. While deep learning models may have advantages over costly laboratory-based experiments and physics-based simulations, existing models have been trained on small datasets with homogeneous demographics and focus on predicting a single output. We developed GaitDynamics, a generative foundation model trained on a large dataset of diverse gait patterns, which allows for flexible inputs, outputs and clinical applications. We illustrate the use of GaitDynamics for: (1) estimating ground reaction forces from kinematics with high accuracy even with missing kinematic data, (2) predicting the effects of gait modifications on knee loading without resource-intensive experiments and (3) predicting kinematic and force changes that occur with increasing running speeds. Our results demonstrate the accuracy and efficiency of GaitDynamics, showing its potential to assess and optimize gait for injury prevention, disease treatment and performance coaching. All data, code and trained models are publicly shared.

Quantifying the dynamics of walking and running, including the motions of the body segments and the joint moments that drive the motion, can help diagnose disease, prevent injury, customize rehabilitation and improve athletic performance. For example, the magnitude, symmetry and variability of gait parameters such as ground reaction forces can provide insights into the mechanisms of sports injury^{1,2} as well as the rehabilitation status of patients with conditions such as stroke^{3,4}, patellofemoral pain⁵ and ligament tears⁶. Measurement of joint loading during walking has also helped develop and personalize gait training interventions to mitigate knee osteoarthritis^{7–10}. Understanding the relationship between running speeds, kinematics and kinetics can enable more effective coaching to optimize human performance^{11,12}.

Traditionally, gait dynamics are measured in specialized laboratories, where trained personnel operate expensive systems to measure

motion and ground reaction forces. The data collection process often requires several hours of expert time for even a single participant, which can be particularly burdensome for participants with mobility limitations. Physics-based simulations, where human dynamics are optimized to match experimental data or hypotheses about the motion, have been applied in many studies to analyse and predict gait dynamics^{13,14}. Previous studies have used simulations to estimate kinetic parameters on the basis of measured kinematics, eliminating the need for force plates that are unavailable outside of specialized laboratories^{15–17}. Simulations have also helped identify the influence of gait modifications on joint moments^{18,19}. Although simulations reduce or eliminate the need for laboratory-based experiments, they can be time-consuming and rely on specialized expertise, often requiring hundreds of hours for a single seconds-long simulation. To mitigate

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Table 1 | Demographics and statistics of training and test-set participants

	# of studies	# of participants (male/female/unknown)	# of trials	Height (m)	Weight (kg)
Training	8	169 (116/44/9)	10,352	1.71±0.09	67.9±12.0
Test	15*	101 (83/17/1)	1,929	1.68±0.16	64.3±16.7

*The 15 studies include 8 studies used for training; however, testing was performed on held-out participants (not used for training) from these studies.

the long computation time, many studies have resorted to simplified human models, for example, constraining the simulation in the sagittal plane to reduce degrees of freedom^{15,20}. The time-consuming nature of human experiments and simulations limit both the widespread assessment of gait parameters and the discovery of effective gait modification strategies across populations who may benefit from such studies.

Many data-driven models have been proposed for estimating gait parameters such as ground reaction forces^{21–25}. While such models can achieve real-time speed and high accuracy, several limitations hinder their applicability in real-world environments. First, these models are designed solely to estimate the unmeasured gait parameters, but none predict how dynamics change due to modifications of specific gait parameters. The ability to predict the dynamic outcomes of modifications (for example, increasing speeds or medial–lateral trunk sway) could potentially enable the discovery of strategies that reduce risky joint loading or enhance performance. Second, the architectures of these models require the inputs to be complete. In practice, the input data can be incomplete due to factors such as infeasibility of sensor placement, data packet drop in wireless transmission and camera occlusion. A flexible model that can process arbitrary combinations of input data would be robust to real-world conditions where inputs might be incomplete. Third, models are typically trained and evaluated on data collected from a single laboratory, which can limit generalizability to other datasets with different cohorts or environments.

In recent years, we have witnessed remarkable advancements in large computer vision and natural language models²⁶. These models are highly flexible in both the inputs they accept and the tasks to which they can be applied, earning them the name, ‘foundation models’. However, since existing human gait dynamics models are primarily trained on small, proprietary datasets and focus on a specific task, foundation models have not yet emerged for human gait dynamics. One major barrier to the development of foundation models is the availability of large-scale datasets with synchronized kinematics and kinetics for gait. The new AddBiomechanics dataset helps overcome this barrier²⁷. The dataset harmonizes movement data from 15 studies, aggregating 34.8 h of data with synchronized kinematics and kinetics from 270 participants (Table 1). The size and diversity of the AddBiomechanics dataset can enable training of a foundation model for gait dynamics that is capable of performing multiple common prediction and analysis tasks.

In this paper, we propose GaitDynamics—a generative foundation model capable of accomplishing multiple common prediction and analysis tasks pivotal to the understanding of gait dynamics. GaitDynamics is trained on the AddBiomechanics dataset, which contains a range of speeds and gait patterns. Our model represents several notable advancements over previous deep learning models in gait analysis. First, the model supports analysis tasks beyond estimating unmeasured gait parameters, such as quantifying how dynamics change with increasing running speeds or modified gait patterns. Second, the model can estimate ground reaction forces on the basis of a flexible combination of input kinematics. Third, the model generalizes to studies not used for training. Notably, despite being trained exclusively on data from healthy participants, GaitDynamics maintains its accuracy among individuals with osteoarthritis. GaitDynamics generates a seconds-long trial within 1 s, which is a thousand-fold more efficient

than physics-based predictive simulation that can take anywhere from 35 minutes to hundreds of hours^{18–20,28–30}. To facilitate broad usage and customizations of GaitDynamics, we have open sourced the trained model and code.

Results

Overview of GaitDynamics

GaitDynamics models full-body kinematics, including pelvis linear velocities with respect to the Earth, joint angles and joint angular velocities, as well as ground reaction forces (hereafter referred to as forces for simplicity) for both feet (Fig. 1). The forces include vertical, anterior–posterior and medial–lateral components, along with the centre of pressure—the point where the force acts on the foot. Unlike previous end-to-end models that map a fixed combination of inputs to outputs, GaitDynamics uses flexible combinations of inputs to generate outputs through a diffusion model. Generated outputs may differ for identical inputs due to diffusion-process randomization, and variability increases when fewer inputs are provided (Supplementary Fig. 5). Model architecture is detailed in Methods. As a foundation model trained on a large training dataset, GaitDynamics can perform multiple downstream tasks. Here we present three of them: (1) estimating forces using flexible combinations of kinematic inputs, (2) predicting the effects of a trunk sway gait modification on risky knee loading without the need for new human experiments and (3) predicting a complex set of spatio-temporal, kinematic and dynamic changes that occur as running speed increases.

Downstream task 1: estimating forces using flexible combinations of kinematic inputs

The ability to accurately infer forces during gait from kinematics would transform biomechanics and healthcare, facilitating applications such as remote rehabilitation, exoskeleton control and running injury prevention. In contrast to previous models that can only use a fixed combination of inputs, GaitDynamics can use flexible combinations of kinematics as inputs (Fig. 2). When using partial-body kinematics as input (for example, without trunk kinematics), GaitDynamics maintained its force estimation accuracy by first using a diffusion model to generate unknown kinematics and subsequently using a refinement model to map full-body kinematics to forces.

The accuracy of GaitDynamics in estimating the vertical and medial–lateral force profiles was significantly higher than that of state-of-the-art models that used a convolutional neural network or a recurrent neural network (Fig. 3, $P \leq 0.01$). The accuracy of GaitDynamics in estimating peak vertical force was significantly higher than that of the recurrent neural network ($P \leq 0.01$) but not that of convolutional neural network ($P = 0.08$). While GaitDynamics had lower anterior–posterior force errors compared with recurrent and convolutional neural network models, there were no significant differences among groups ($P = 0.26$). When using full-body kinematic inputs, the mean absolute errors of GaitDynamics in estimating the vertical force profile, anterior–posterior force profile, medial–lateral force profile and peak vertical force were 3.2%, 1.2%, 0.7% and 3.7% of body weight (BW), respectively. Further, the errors of GaitDynamics in estimating the peak vertical force were smaller than clinically meaningful thresholds—minimal detectable change (MDC)—for both healthy participants (MDCs of 10.2–13.0% of BW)^{31,32} and post-stroke individuals (MDC of 4.7% of BW)³³. This demonstrates that GaitDynamics can identify changes in vertical forces that are greater than measurement errors³⁴.

For studies with healthy participants and not used to train the model, mean absolute error ranges were 1.1–3.9%, 0.8–1.5% and 0.5–0.7% of BW for vertical, anterior–posterior and medial–lateral force profiles, respectively, comparable to those of held-out participants from the training set studies (1.3–3.3%, 0.4–1.5% and 0.3–2.0% of BW; Supplementary Table 1). Although we only used healthy participants’ data to train GaitDynamics, the model maintained its accuracy for

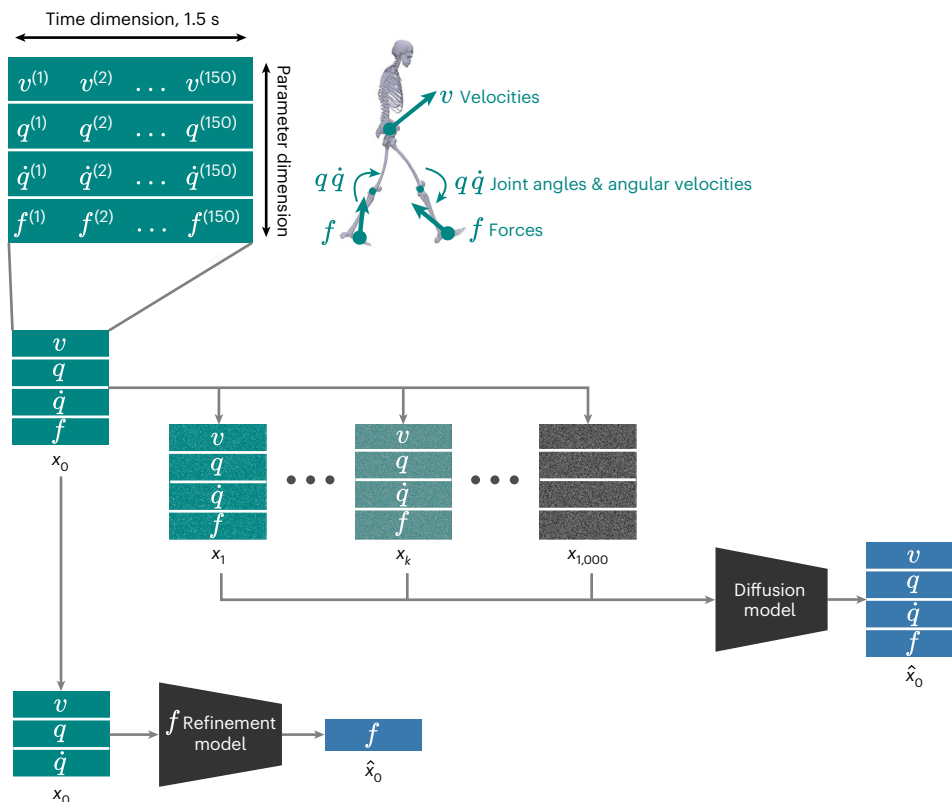


Fig. 1 | The data window and training of GaitDynamics. The data comprise a 2D window with a 1.5-s time dimension and a parameter dimension. The parameter dimension includes pelvis linear velocities v , joint angles q , joint angular velocities $\dot{q}$ and forces f . GaitDynamics has one diffusion model and one force refinement model. For the diffusion model, we followed the definition of diffusion as a Markov chain that gradually adds Gaussian noise to the clean windows on the basis of step k . At $k = 0$, x_0 is a clean window with no noise

added, whereas at $k = 1,000$, $x_{1,000}$ is pure Gaussian noise. A diffusion transformer was trained to recursively restore the clean windows from the windows with different amounts of noise. While forces are part of the diffusion model parameters, we found that a subsequent end-to-end model achieved higher force prediction accuracy when using full-body kinematics (v , q and $\dot{q}$) as inputs. Therefore, we separately trained an end-to-end model, which we refer to as the force refinement model.

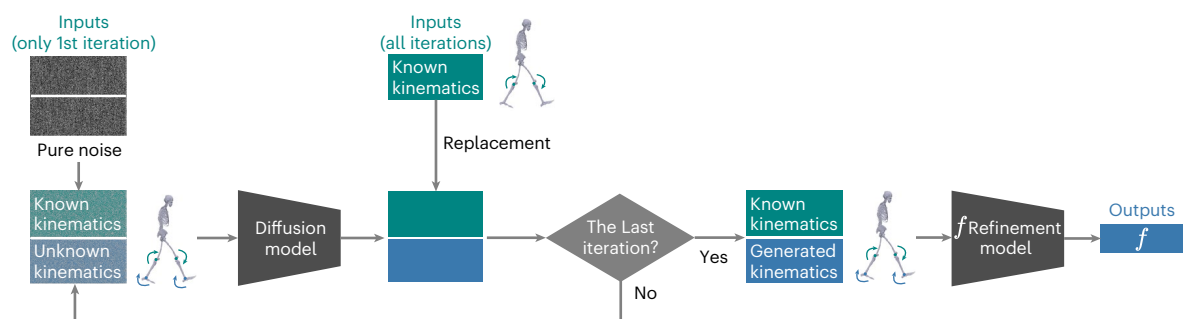


Fig. 2 | Force estimation using GaitDynamics with partial-body kinematics as inputs. GaitDynamics generates unknown kinematics using inpainting, where the diffusion model recursively denoises the window on the basis of the known portion. Inpainting begins by using the diffusion model to denoise pure Gaussian noise. The unknown portion of the denoised window is forward diffused to match the noise level of the next iteration, whereas the known portion of the denoised

window is replaced by forward-diffused input kinematics. The same denoising, diffusion and known portion replacement processes are iteratively applied to the window for a total of 50 iterations, where the quality of generation gradually increases. Finally, the unknown kinematics generated in the last iteration as well as known kinematics are fed into the refinement model to estimate forces.

individuals with osteoarthritis, whose average mean absolute errors were 3.8%, 1.6% and 0.9% for vertical, anterior–posterior and medial–lateral force profiles, respectively. However, the accuracy was lower for a post-stroke individual whose average mean absolute errors were 10.1%, 3.1% and 1.0% of BW for vertical, anterior–posterior and medial–lateral force profiles.

To investigate the influence of incomplete kinematic inputs on the accuracy of estimating forces, we iteratively masked the kinematics of

the trunk, pelvis, hip, knee and ankle joints from the inputs. GaitDynamics fills the masked kinematics through a process called inpainting, where the unknown portions of the kinematics are generated on the basis of the known portions. Compared to using full-body kinematics as inputs, using partial-body kinematics only slightly decreased the accuracy of GaitDynamics by less than 2.2% of BW, depending on the joint kinematics that were masked (Fig. 4). The largest accuracy decrease of 2.2% of BW occurred when kinematics of both the left and right

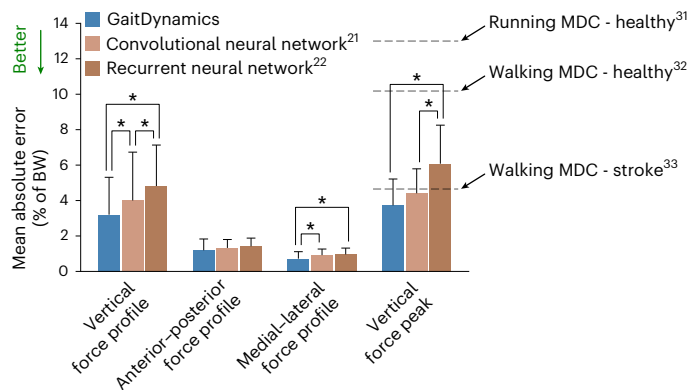


Fig. 3 | Mean absolute errors of force estimation for the GaitDynamics model and recently published data-driven models when using full-body kinematic inputs. Peak vertical force was computed as the maximum value of the vertical force profile during each gait cycle. Mean $\pm$ 1 s.d. of mean absolute errors (% of BW) across 14 test-set studies ($n = 14$). The horizontal dashed lines are the MDCs reported in literature. An asterisk, *, denotes significant difference. We used the non-parametric Friedman test to assess whether there were significant differences for each force variable. When the Friedman test was statistically significant ($P \leq 0.05$), we conducted post-hoc pairwise comparisons testing with the two-sided Wilcoxon signed-rank test and Bonferroni correction to account for multiple comparisons ($\alpha = 0.017$).

hip were masked. When the trunk, pelvis, both knees or both ankles (including subtalar joints) were masked, the accuracy changes were only 0.1%, 1.5%, 0% and 0.1% of BW, respectively. The larger accuracy decrease when masking both hips' kinematics can be attributed to the greater number of degrees of freedom at the hip, which is 3, compared with 1 to 2 degrees of freedom for knee or ankle. We also compared the performance of GaitDynamics against a baseline method of accounting for missing kinematics, median filling, where we used the median values of each degree of freedom to fill unknown input kinematics and then used the same refinement model to estimate forces. Compared to using full-body kinematics, filling kinematics of the trunk, pelvis, hip, knee or ankle with their median values decreased the accuracies by 3.8%, 36.4%, 33.4%, 14.1% and 3.9% of BW, substantially larger than those obtained by the inpainting of GaitDynamics.

We used heuristics to prune noisy, erroneous and unusable data, which accounted for ~25% of trials of the dataset. To evaluate the benefits of the pruning, we compared model performance when trained on all the data versus the data after pruning. Compared to training with all the data, training with data after pruning slightly reduced mean absolute errors from 4.08% to 3.73% of BW for the peak vertical force, from 3.30% to 3.20% of BW for the vertical force profile, from 1.23% to 1.19% of BW for the anterior-posterior force profile, and from 0.84% to 0.72% of BW for the medial-lateral force profile.

Downstream task 2: predicting knee adduction moment during trunk sway gait without experiments

Previous controlled experiments identified that increasing medial-lateral trunk sway during walking^{35,36} can reduce knee adduction moments, which are associated with medial compartment knee osteoarthritis^{37–39}. We used the GaitDynamics model to replicate the relationship between trunk sway gait and knee loads, without needing experiments (Fig. 5). To perform experiment-free prediction, we first used normal walking trials from a test-set study⁴⁰ to create synthetic trunk sway gait, increasing medial-lateral trunk angles by $2.0\times$, $2.5\times$ and $3.0\times$ of their original magnitudes. We selected $2.0\times$ to $3.0\times$ to encompass values both smaller and larger than the experimental large trunk sway condition (Fig. 6). Then, we inputted synthetic trunk kinematics into the GaitDynamics model to generate full-body kinematics and forces of trunk sway gait.

We used the normal walking data to guide the model in generating walking gaits through guided diffusion, a method that uses data or constraints to regularize the generation process of diffusion models^{41–43}. For this downstream task, we used joint angles of normal walking to regularize joint angles of GaitDynamics generations if they deviated by more than 10% of the range of motion. Finally, we computed the knee adduction moment using the cross product of the force vector and the lever arm vector from the knee joint centre to the centre of pressure.

The knee adduction moments predicted by GaitDynamics were consistently lower as the synthetic trunk sway angles increased from $2\times$ to $3\times$ of normal trunk sway (Fig. 6), which is consistent with previous experiments⁴⁰. The peak trunk sway angles were 13° for experimental normal walking, 28° for experimental large trunk sway gait, and $25\text{--}38^\circ$ for synthetic large trunk sway gait. The first peak of knee adduction moment reduced by 1.3% of body weight (BW) $\times$ body height (BH) in experimental measurements, which was larger than the $0.5\text{--}1.1\%$ of BW $\times$ BH reductions in synthetic trunk sway gaits generated by GaitDynamics. The second peak of knee adduction moment reduced by 0.8% of BW $\times$ BH in experimental measurements, which was smaller than the $0.9\text{--}1.0\%$ of BW $\times$ BH reductions in synthetic trunk sway gaits generated by GaitDynamics. It took less than 1 s to generate a 1.5-s walking trial on a single Nvidia RTX3050 Laptop graphics processing unit (GPU).

Downstream task 3: gait generation at synthetically altered running speeds

Increasing gait speed is a common goal in both rehabilitation and athletics. GaitDynamics can also predict the changes in stride length, peak knee flexion and peak forces with altered running speeds. We created synthetic running velocities ranging from 3 m s^{-1} to 5 m s^{-1} by linearly scaling the anterior-posterior component of pelvis linear velocities measured during 4 m s^{-1} running⁴⁴ to values between 75% and 125%. Then, we inputted the scaled anterior-posterior component of pelvis linear velocities along with the original medial-lateral and vertical components of pelvis linear velocities into the GaitDynamics model to generate full-body kinematics and forces at synthetic running speeds. During generation, we used guided diffusion to regularize any generated joint angles that deviated from the 4 m s^{-1} experimental data by more than 30% of each joint's range of motion.

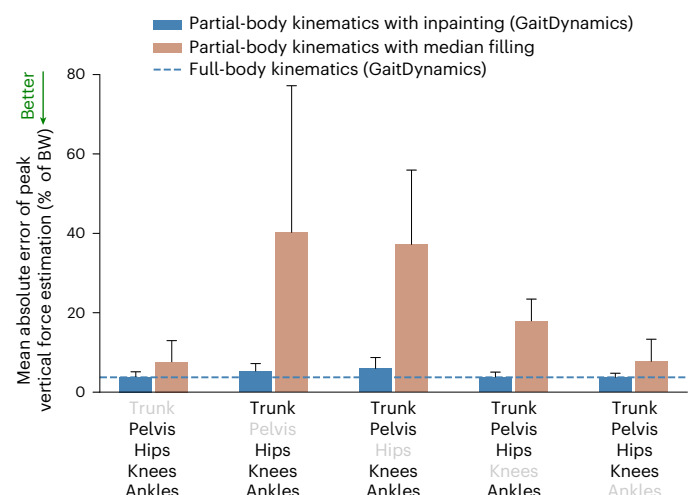


Fig. 4 | Mean absolute error of peak vertical force estimation when using various combinations of partial-body kinematics as inputs. The light grey text denotes the masked (excluded) kinematic inputs. Mean $\pm$ 1 s.d. across 14 test-set studies. The dashed line is the mean absolute error of GaitDynamics when using full-body kinematics as input. The accuracy of inpainted kinematics and conditions with more kinematic input combinations being masked can be found in Supplementary Table 3, which demonstrates that accuracy decreases were smaller for combinations that included the kinematics of both pelvis and hips.

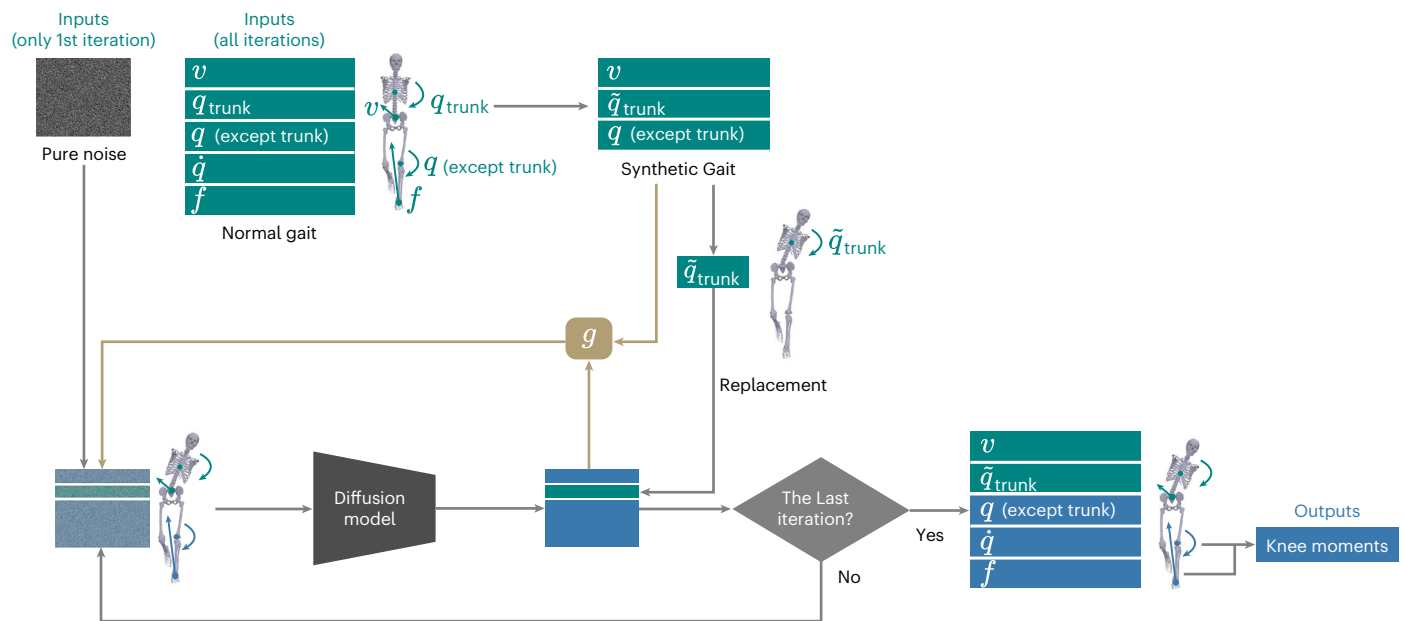


Fig. 5 | Experiment-free prediction of knee adduction moment during walking with large medial–lateral trunk sway angles. The process is similar to the inpainting described in Fig. 2, with two differences. First, instead of using experimental data as input, synthetic trunk kinematics with increased medial–lateral trunk sway are used as inputs. Second, a loss function, g , is used to regularize large discrepancies in v and q between synthetic gait and model

generations for each iteration. Discrepancies larger than a threshold are back propagated to refine the noisy motion using gradient descent optimization. Finally, on the basis of the generated q (except for trunk) and f , we compute the knee adduction moment using the cross product of the force vector and the lever arm vector from the knee joint centre to the centre of pressure.

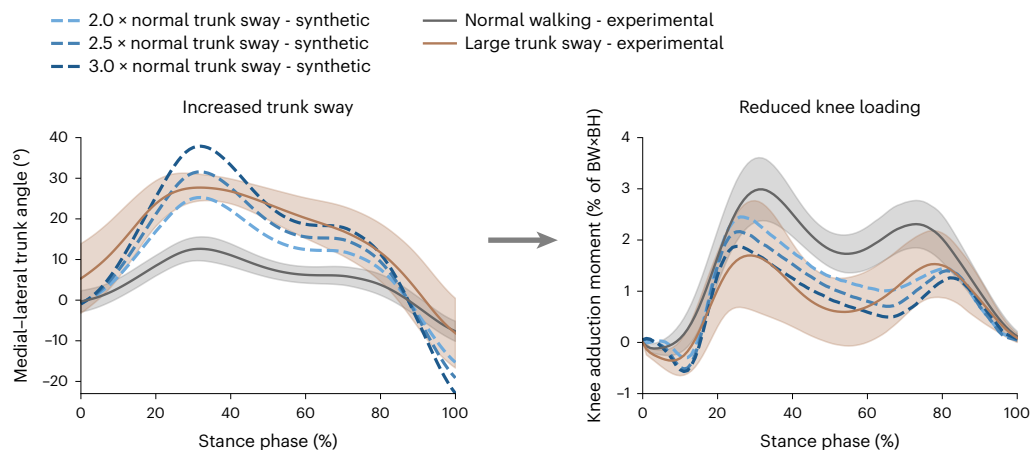


Fig. 6 | Medial–lateral trunk angles and their corresponding knee adduction moments. Angles and moments are from normal gait⁴⁰ (grey), experimental large trunk sway gait⁴⁰ (brown) and synthetic large trunk sway gait (blue). The shaded areas represent 1 s.d. of experimental knee adduction moment. With the

increase of synthetic trunk sway angles, the predicted knee adduction moment consistently decreased. The 1 s.d. of synthetic trunk sway gait as well as synthetic 1.0× normal trunk sway can be found in Supplementary Fig. 3.

As running speed increased, GaitDynamics consistently predicted greater stride lengths, peak knee flexion angles, peak vertical forces and anterior–posterior forces, following the same trend as experimental measurements⁴⁴ (Fig. 7). As running speed increased from 3 m s^{−1} to 5 m s^{−1}, there was an expected large increase in the stride length (1.8 m to 3.0 m for GaitDynamics generations vs 2.0 m to 2.7 m for experimental measurements), as well as a large increase in peak knee flexion angle (109° to 121° for GaitDynamics generations vs 100° to 125° for experimental measurements). For ground reaction forces, as the running speed increased from 3 m s^{−1} to 5 m s^{−1}, there was an expected large increase in the peak anterior–posterior force (39% for GaitDynamics generations vs 48% for experimental measurements), as well as a much smaller increase in peak vertical force (8% for GaitDynamics generations

vs 10% for experimental measurements). Similar to the walking generations described above, it took less than 1 s to generate dynamics of a 1.5-s running trial on a single Nvidia RTX3050 Laptop GPU.

In contrast to GaitDynamics, the state-of-the-art motion generation model, OmniControl, is only capable of generating kinematics parameters⁴⁵. Moreover, unlike the experimental measurements, the stride length and peak knee flexion generated by OmniControl did not consistently increase at higher synthetic running speeds. In addition, we manually reviewed the OmniControl generations and found that 3 out of 10 generations at a speed of 5 m s^{−1} were unsuccessful, as they did not exhibit the characteristics of human running gait (for example, nearly static posture over time). We removed the three unsuccessful generations from analysis.

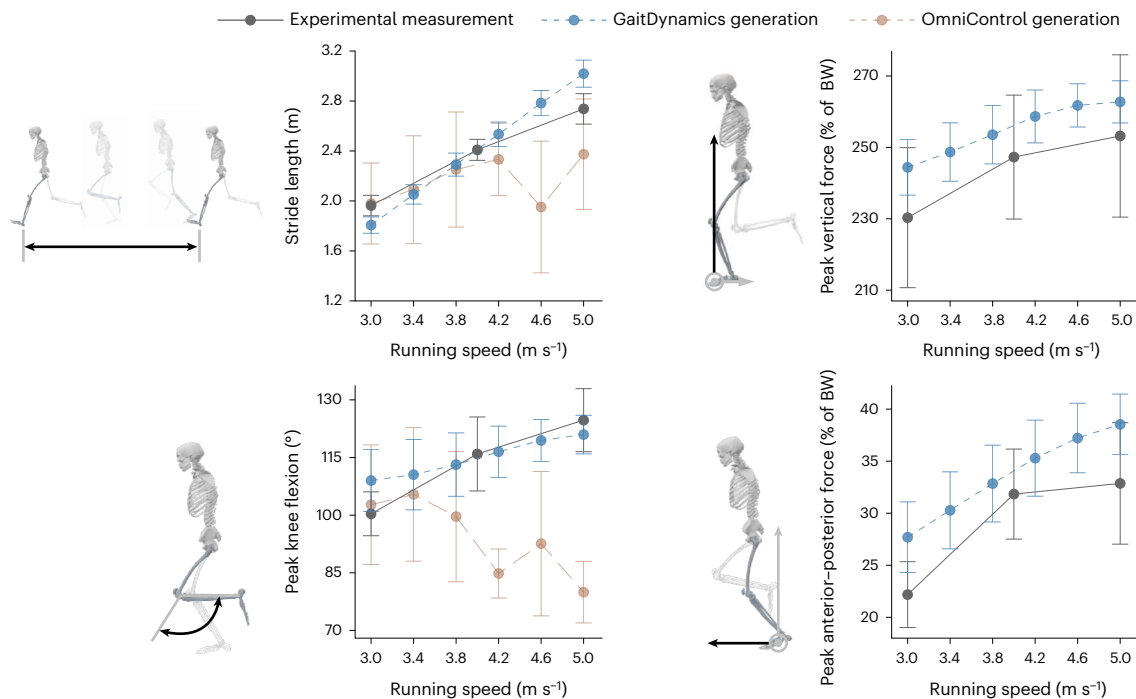


Fig. 7 | Gait parameters across running speeds. The parameters were measured in experiments⁴⁴ (grey), generated by GaitDynamics (blue), and generated by a state-of-the-art motion generation model, OmniControl⁴⁵ (orange). Mean $\pm$ 1 s.d. for 10 participants. Of 10 OmniControl generations at a speed of 5 m s⁻¹, 3 were unsuccessful and thus excluded from the mean and standard deviation

calculations. Both kinematic and kinetic parameters generated by GaitDynamics consistently increased with running speed, and their slopes were aligned with those of experimental measurements. In contrast, OmniControl can only generate kinematic parameters.

Discussion

We developed the first generative foundation model to quantify gait dynamics and analyse their changes across various gait patterns. GaitDynamics uses a diffusion model, which was able to learn a robust and meaningful representation of gait dynamics encapsulated in the training data. While many existing deep learning models have been proposed for a single gait estimation task, GaitDynamics allows flexible input combinations while maintaining accuracy, positioning it as a foundational model for multiple downstream tasks beyond estimating unknown parameters. We show the utility of GaitDynamics on three diverse downstream tasks, where it successfully predicted (1) forces that drive the movement with full- or partial-body kinematics, (2) the influence of gait modifications on risky knee loading without the need for costly human experiments and (3) comprehensive kinematic and kinetic changes at a range of gait speeds.

In contrast to previous models developed on small, proprietary datasets, we trained and evaluated GaitDynamics using data from 15 previous studies, aggregating more than 34.8 h of synchronized kinematics and force data at various gait speeds and patterns. The extensive size of this dataset enabled GaitDynamics to generalize to unseen studies which were not explicitly included in the training data. Although the dataset used for developing GaitDynamics is harmonized from high-quality biomechanical studies, noisy and erroneous data still exist in some trials. We used heuristics to prune such data, which helped improve the accuracy of force estimation. Thus, when using open-source datasets to train foundation models for human movement analysis, improving data quality can even outweigh increasing the quantity of data. This observation aligns with practices in contemporary training methods for large language models^{46,47}.

For ground reaction force estimation, GaitDynamics achieved smaller errors than state-of-the-art models and was generalizable to studies and participants not used for training (Supplementary Table 1). The training set studies exclusively included healthy participants, so

GaitDynamics was not exposed to pathological gait during training. Still, the model maintained its accuracy for individuals with osteoarthritis, although it showed a drop in accuracy for a post-stroke individual. One explanation for the accuracy drop difference is that the post-stroke individual has a substantially lower cadence compared with healthy participants, whereas individuals with osteoarthritis maintained cadences similar to those of healthy participants.

For data longer than 1.5 s, we generated overlapping windows and used the mean prediction; the predictions were continuous over time (Supplementary Fig. 1). We also investigated whether the model could still make continuous predictions by generating non-overlapping windows. We found that the predictions were extremely close to the mean predictions from overlapping windows, except for subtle discontinuities in a few locations (Supplementary Fig. 2). Thus, for tasks with strict latency requirements, users of GaitDynamics may consider generating non-overlapping windows to enable faster prediction.

GaitDynamics demonstrated the feasibility of generating forces and joint moments corresponding to synthetic kinematic data altered from normal gait, enabling experiment-free, efficient prediction of gait modification outcomes. GaitDynamics's ability to predict knee joint moments during trunk sway gait is probably learned from one of the training set studies involving participants walking with varied trunk sway angles⁴⁸. Although the training set participants performed bilateral trunk sway on a treadmill, GaitDynamics successfully predicted knee adduction moment reductions that aligned with those of test-set participants performing ipsilateral trunk sway overground.

GaitDynamics can predict interactions between pairs of gait parameters (for example, joint angle vs gait speed), potentially making it a valuable tool for more effective coaching and rehabilitation. For example, it can identify parameters that are outside of the distribution of healthy gait, thereby training the user to walk and run more naturally. The kinematic and kinetic parameters generated by GaitDynamics aligned with those of experimental measurements

across 3–5 m s⁻¹ running. In contrast, a state-of-the-art motion generation model from the computer graphics community, OmniControl, cannot generate kinetic parameters or predict the changes in kinematic parameters at increased running speeds. However, GaitDynamics may not generate running gaits at speeds greater than 5 m s⁻¹, since the highest running speed included in its training set was 4.7 m s⁻¹ (ref. 49) (Supplementary Fig. 4). Moreover, one limitation of our guided diffusion formulation is that it constrains temporal scaling or shifts in the generated gait relative to the 4 m s⁻¹ experimental data. As a result, GaitDynamics cannot predict large changes in temporal parameters such as cadence, which is a critical measure of gait performance and health. To better predict changes in cadence, instead of costly model retraining or fine tuning, future studies may customize the loss function g (equation 13) of the guided diffusion to allow for temporal scaling and shifts in the generated gait.

GaitDynamics can generate a 1.5-s data window within 1 s on a personal computer, making it substantially more efficient than physics-based predictive simulations. Recent simulation studies have attempted to accelerate the simulation speed using techniques such as direct collocation and algorithmic differentiation, reducing the simulation time from hundreds of hours^{18,19} to ~1 h (refs. 28,29,50). Still, GaitDynamics is a thousand-fold more efficient than the accelerated simulation, and thus may enable systematic analyses that were previously infeasible. For example, it may enable systematic investigation of various gait modification strategies or their combinations at a fine-grained resolution, leading to the discovery of new gait modification strategies that maximize clinical outcomes.

Our study has a few limitations. First, GaitDynamics cannot model walking and running turns because we pruned trials where the forward moving direction changed by more than 45°. Second, the kinematic inputs of this study were limited to marker-based motion capture data; therefore, it is unclear whether GaitDynamics can maintain its performance using kinematics estimated by sensors such as inertial measurement units and smartphone cameras. Our open-source code and model parameters enable future work across the broader research community to evaluate the robustness and accuracy of GaitDynamics using kinematic inputs from these sensors. Third, as suggested by the name, GaitDynamics may not be applied to non-gait prediction and analysis tasks because its training data was predominantly walking and running gait. Future work may consider incorporating more non-gait data through human experiments, by self-supervised learning methods that do not require force measurements^{51–53}, or by simulations that do not rely on human experiments^{54,55}. Future foundation models could also incorporate muscle activation to provide muscle-level insights as an alternative to computationally intensive musculoskeletal simulations.

This study marks a notable advancement in human gait analysis, transitioning away from resource-intensive laboratory-based experiments and simulations. This work will advance the understanding of fundamental principles that govern human movement, potentially facilitating injury prevention, movement rehabilitation, exoskeleton developments and sports coaching.

Methods

Datasets

We trained and evaluated our model using the AddBiomechanics dataset²⁷ that harmonized marker-based motion capture and force plate data from 15 previous studies^{21,40,44,48,56–66}. The data were processed by the AddBiomechanics engine⁶⁷, which used marker data to perform scaling of an OpenSim skeletal model without arms^{68,69} and inverse kinematics to determine the pelvis (root of the skeletal model) positions and joint angles over time. Note that we did not use the force-based kinematics tuning described in the AddBiomechanics paper to prevent the leakage of force data (output of GaitDynamics) into the kinematics (input of GaitDynamics).

We performed the following pre-processing steps to further process the data before using them for model training.

- Of the studies in the AddBiomechanics dataset, 8 out of 15 studies provided kinematics at 100 Hz, whereas the remaining 7 studies provided kinematics between 120 and 250 Hz. To minimize distribution discrepancies, we linearly resampled the data from these 7 studies to 100 Hz.
- We computed joint angular velocities for all the joints of the OpenSim skeletal model, including 6 one-dimensional (1D) joints, that is, both knees, ankles and subtalar, as well as four 3D joints, that is, trunk, pelvis and both hips. For 1D joints, we used B-splines to fit the angles and then took the first derivatives of the splines as angular velocities⁷⁰. For 3D joints, we converted Euler angles to a rotation matrix (R) for each time (t) and then used finite difference to compute a 3×3 angular velocity matrix ($\tilde{\omega}_{3 \times 3}$) for extracting angular velocities (ω) as follows⁷¹:

$$\tilde{\omega}_{3 \times 3} = \begin{bmatrix} \tilde{\omega}_{11} & \tilde{\omega}_{12} & \tilde{\omega}_{13} \\ \tilde{\omega}_{21} & \tilde{\omega}_{22} & \tilde{\omega}_{23} \\ \tilde{\omega}_{31} & \tilde{\omega}_{32} & \tilde{\omega}_{33} \end{bmatrix} = \frac{R(t+1) - R(t-1)}{2 \times \Delta t} \quad (1)$$

$$\omega = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = 0.5 \times \begin{bmatrix} \tilde{\omega}_{32} - \tilde{\omega}_{23} \\ \tilde{\omega}_{13} - \tilde{\omega}_{31} \\ \tilde{\omega}_{21} - \tilde{\omega}_{12} \end{bmatrix} \quad (2)$$

- Euler angles provided by the AddBiomechanics dataset are discontinuous representations of orientation that are more difficult for deep neural networks to learn compared with continuous representations⁷². Thus, we first converted all the Euler angles (that is, trunk, pelvis and both hips) to 3×3 rotation matrices and then extracted the first two rows of each matrix to form a 6D vector.
- For each trial, we applied a rotation matrix that converts the median moving direction (yaw angle of the pelvis) to zero, to the moving direction of all frames. This made all gait trials independent of the direction in which the skeletal model is facing.
- The dataset included both overground and treadmill gaits, where gait velocities should be measured relative to the Earth and the treadmill belt, respectively. To mitigate this discrepancy, for overground trials, we computed the pelvis linear velocities by finite differencing of the positions. For treadmill trials, we first estimated the belt speed by finite differencing of anterior–posterior foot position during mid-stance where the vertical force is larger than 40% of body weight. Then, the missing belt speeds between mid-stances were estimated via linear interpolation. Finally, the pelvis linear velocities were estimated as follows:

$$\begin{aligned} \hat{v}_x &= v_{\text{pelvis},x} - (\hat{v}_{\text{right belt}} + \hat{v}_{\text{left belt}})/2 \\ \hat{v}_y &= v_{\text{pelvis},y} \\ \hat{v}_z &= v_{\text{pelvis},z} \end{aligned} \quad (3)$$

where v_{pelvis} denotes the ground-truth pelvis linear velocities in the Earth frame, while $\hat{v}_{\text{right belt}}$ and $\hat{v}_{\text{left belt}}$ denote the estimated belt speeds.

- We expressed the centre of pressure in the calcaneus frame of the stance foot rather than in the Earth frame. Moreover, when the vertical force is small, noise in the force measurements can cause centre of pressure drifts. Therefore, we divided the centre of pressure by the height and multiplied it by the magnitude of the vertical force.
- We also performed the following quality check steps to prune noisy, erroneous, or unusable data.
- Due to the non-convex nature of inverse kinematics, a few trials had implausible trunk rotation angles (yaw angle) that

- were consistently larger than 45° . We pruned 121 trials (0.5%) whose average trunk rotation angles were larger than 45° .
- We aligned the direction in which the skeletal model is moving, as discussed above. However, alignment was infeasible for 1,682 trials (7.5%) because the range of moving direction (yaw angles of the pelvis) exceeded 45° . Therefore, we pruned these trials.
 - Trial lengths were computed as the time period for which both kinematics and ground reaction force measurements were available. Trials shorter than the minimal length of the model input, 1.5 s, cannot be used for model training. We pruned 3,669 trials (16.4%) that were shorter than 1.5 s.
 - We noticed that the distances between the centre of pressure and the contact foot were unfeasibly large in a few component studies of the dataset. We pruned 3,725 trials (16.7%) where this distance exceeded 0.3 m for more than 0.2% of the stance phase.
 - For each frame, if the angular velocity of any joint was higher than $2,000^\circ \text{ s}^{-1}$, this frame, its previous 2 frames and its subsequent 2 frames were pruned from the dataset. The threshold, $2,000^\circ \text{ s}^{-1}$, is a conservative value determined on the basis of maximum hip and knee angular velocities ($\leq 800^\circ \text{ s}^{-1}$) in a high-speed (6.5 m s^{-1}) running study⁷³. There were 833 frames that exceeded this threshold, accounting for fewer than 0.01% of the total number of frames.

This data pruning process yielded 34.8 h of data from 270 participants. For model training, we included both gait and non-gait motions such as squat, hopping and idling. We trained our model using 8 of the 15 studies^{48,56–61,66} in the AddBiomechanics dataset, including all participants ($n = 169$) except one from each study for training (Table 1). Note that we completely held out the remaining 7 studies^{21,40,44,62–65} from the training process. Thus, we evaluated the model's performance on participants ($n = 101$) from two groups: the 8 held-out participants from the studies used for training, as well as all participants from the 7 studies not used for training. The 8 held-out participants followed the same training and test-set split as in the AddBiomechanics dataset paper²⁷. The two test groups allowed us to evaluate the model's generalizability in force estimation across both studies included and not included in the training process. To evaluate the benefits of the proposed data pruning criteria, we compared model performance when trained on data after pruning versus on all the data (except for trials shorter than 1.5 s that are not usable).

GaitDynamics architecture and training

GaitDynamics models full-body kinematics, including pelvis linear velocities with respect to the Earth (v), joint angles (q) and joint angular velocities ($\dot{q}$), as well as ground reaction forces and centres of pressure for both feet (f). These parameters were concatenated into 2D data windows, x , which have one parameter dimension and one temporal dimension (Fig. 1). The parameter dimension includes 63 components: (1) 3 components for pelvis linear velocities in the anterior–posterior, medial–lateral and vertical directions; (2) 6 components for 1D joint angles of both knees, ankles and subtalar; (3) 24 components for 6D vectors of trunk, pelvis and both hips; (4) 6 components for 1D angular velocities of both knees, ankles and subtalar; (5) 12 components for 3D angular velocities of trunk, pelvis and both hips; and (6) 12 components for ground reaction forces and centres of pressure of both feet in the anterior–posterior, medial–lateral and vertical directions. Each parameter was normalized by min–max normalization whose bias and scale were determined by the minimum and maximum values of the entire training set data. The temporal dimension consists of 150 time steps, corresponding to 1.5 s of data. This window length is conservatively chosen to exceed durations of typical gait cycles, allowing our model to access all phases of gait cycles for making predictions.

GaitDynamics consists of one diffusion model and one force refinement model. The diffusion model learns to estimate the data

distribution of the human gait using iterative processes of adding noise to the window and denoising the windows. It has a transformer architecture similar to those of previous human kinematics modelling studies^{74,75}. During training, clean windows were diffused into noisy windows (Fig. 1) through a Markov noising process⁷⁶ as follows:

$$Q(x_k, |, x_0) \sim N(\sqrt{a_k}x_0, (1 - a_k)I) \quad (4)$$

where $k \in [0, 1, 000]$ is the diffusion step, $Q(x_k|x_0)$ denotes the probability distribution of noisy window (x_k) conditioned on the clean window (x_0), N denotes a normal distribution with a mean value of $\sqrt{a_k}x_0$ and a covariance matrix of $(1 - a_k)I$, $\sqrt{a_k}$ are constants following a monotonically decreasing (cosine) schedule for increasing k , and I denotes an identity matrix. At $k = 0$, x_0 is a clean window with no noise added, whereas at $k = 1, 000$, the window $x_{1,000}$ is pure Gaussian noise. The diffusion model is trained to directly predict clean windows using x_k and k as inputs, with mean-squared-error loss function $L_{\text{diffusion}}$:

$$\hat{x}_0 = \epsilon_\theta(x_k, k) \quad (5)$$

$$L_{\text{diffusion}} = |x_0 - \hat{x}_0|^2 \quad (6)$$

where $\hat{x}_0$ denotes the predicted clean window, ϵ_θ denotes the diffusion model, and θ denotes trainable parameters of the model. The transformer of the diffusion model has 4 self-attention blocks, each configured with 4 attention heads, a 256-dimensional latent space and 1,024 feedforward units. We used rotary position embedding⁷⁷ along the temporal dimension, 10% dropout, 10^{-5} layer normalization and rectified linear unit (ReLU) activation function. We used the Adan optimizer⁷⁸ with a learning rate of 10^{-3} , default β ($\beta_1 = 0.02$, $\beta_2 = 0.08$, $\beta_3 = 0.01$) and a minibatch size of 256 for 1×10^5 backpropagation steps. When sampling windows for each minibatch, we implemented uniform probability sampling across all training set studies and trials to avoid biases from training set studies with larger sizes or from longer trials.

While the diffusion model can estimate ground reaction forces through inpainting, a force refinement model that uses full-body kinematics (v, q and $\dot{q}$) as inputs to predict ground reaction force outputs (f) can achieve higher accuracy (Supplementary Table 2). The force refinement model has a transformer architecture similar to that of a previous kinematics-based force estimation study⁵³. The transformer has 6 self-attention blocks, each configured with 8 attention heads and 512 feedforward units. We used the same rotary position embedding⁷⁷ along the temporal dimension, 10% dropout, 10^{-5} layer normalization, rectified linear unit (ReLU) activation function and mean-squared-error loss function $L_{\text{refinement}}$:

$$L_{\text{refinement}} = |f_0 - \hat{f}_0|^2 \quad (7)$$

We used the same Adan optimizer⁷⁸ with a learning rate of 10^{-3} , default β ($\beta_1 = 0.02$, $\beta_2 = 0.08$, $\beta_3 = 0.01$) and a minibatch size of 256 for 3×10^5 backpropagation steps. When sampling windows for each minibatch, we applied the same uniform probability sampling across all training set studies and trials. GaitDynamics was implemented in Python 3.9 with PyTorch 1.13 and Numpy 1.23. It took 30 h on a single NVIDIA RTX A6000 GPU to sequentially train a diffusion model followed by a force refinement model.

Evaluation

We evaluated GaitDynamics on three downstream tasks: (1) force estimation using various input combinations, (2) knee adduction moment prediction using synthetically increased trunk sway angles and (3) kinematic and kinetic prediction using synthetic running velocities. We evaluated Downstream task 1 on the entire test set, including all the participants from studies not used for training as well as all held-out

participants from the studies used for training. For Downstream tasks 2 and 3, we used two studies not used for training. These two studies were the only ones containing experimental measurements for comparison with GaitDynamics generations.

Downstream task 1: force estimation using flexible combinations of kinematic inputs. For partial-body kinematics, GaitDynamics first uses the diffusion model to generate unknown kinematics, followed by a refinement model that maps full-body kinematics to forces (Fig. 2). The diffusion model generates unknown kinematics using the Denoising Diffusion Implicit Models (DDIM) sampling⁷⁹ through a process called inpainting⁸⁰, where the unknown portion of the window is recursively denoised on the basis of the known portion. Specifically, the diffusion starts by initializing the noisy window x_k as pure Gaussian noise and denoising it on the basis of the current diffusion step k ($k = 1, 000$ for the first iteration) as follows:

$$\hat{x}_0 = \epsilon_\theta(x_k, k) \quad (8)$$

where $\hat{x}_0$ denotes the denoised window of the current iteration and ϵ_θ denotes the diffusion model. Next, the denoised window is diffused to create the noisy window of the next iteration (x_{k-s}) as follows:

$$\hat{\epsilon} = \frac{x_k - \sqrt{\alpha_k} \hat{x}_0}{\sqrt{1 - \alpha_k}} \quad (9)$$

$$x_{k-s} = \sqrt{\alpha_{k-s}} \hat{x}_0 + \sqrt{1 - \alpha_{k-s}} \hat{\epsilon} \quad (10)$$

where $\hat{\epsilon}$ denotes the predicted noise at step k , α_k denotes a constant following the noise scheduler, and s denotes the step size for DDIM sampling, which is set to 20 throughout this study. Next, to constrain the known portion of window to the input kinematics, we forward diffused the input kinematics and used it to replace the known portion of x_{k-s} as follows:

$$x_{k-s}^{\text{input}} \sim N(\sqrt{\alpha_{k-s}} x_0^{\text{input}}, (1 - \alpha_{k-s}) I) \quad (11)$$

$$x_{k-s} = m \odot x_{k-s}^{\text{input}} + (1 - m) \odot x_{k-s} \quad (12)$$

where x_0^{input} denotes a data window with the input kinematics assigned to the known portion and zeros assigned to the unknown portion, $\odot$ denotes element-wise product, and m denotes a binary mask, with ones assigned to the known portion and zeros to the unknown portion of the data window. Note that we simply set the unknown portion of x_0^{input} to zero, since they were masked out by m and therefore would not influence the results. The denoising, diffusion and known portion replacement processes were recursively applied to the window for 50 iterations (that is, 1,000 steps over a step size of 20), after which the inpainting process was completed and the unknown kinematics was generated. Finally, the generated kinematics along with the known kinematics were provided as inputs into the refinement model for force estimation.

We used a sliding window with a length of 1.5 s and a step size of 0.15 s to cover entire trials longer than 1.5 s. For the last window, if a 0.15-s step size would exceed the remaining trial length, we adjusted the step size to match the remaining length. For overlapping portions of windows, we used the mean value of all the predictions as the final prediction. The accuracy of force estimation across the 14 studies in the AddBiomechanics dataset is not normally distributed (Supplementary Table 1). Thus, we used the non-parametric Friedman test to assess whether there were significant differences between GaitDynamics and two baseline neural network models for each force variable. When the Friedman test was statistically significant ($P \leq 0.05$), we conducted post-hoc pairwise comparisons testing with the

two-sided Wilcoxon signed-rank test and Bonferroni correction to account for multiple comparisons ($\alpha = 0.017$).

To evaluate various combinations of partial-body kinematics, we iteratively masked kinematics of trunk, pelvis, hip, knee and ankle joints from inputs, used inpainting to generate masked kinematics and estimated forces using the refinement model. We also implemented a baseline method for filling masked kinematics called median filling, where the unknown kinematics are filled by its median value. Median filling used the same force refinement model as inpainting for force estimation.

We compared the performance of GaitDynamics in force estimation against two state-of-the-art models specifically designed to use joint kinematics for force estimation. Their architectures—convolutional neural network²¹ and recurrent neural network²²—are representative architectures distinct from the transformers used in GaitDynamics. For both models, we implemented their model architectures and used the same dataset, optimizer, batch size and learning rate as those of GaitDynamics. On the basis of the lowest training set loss, we trained the convolutional and recurrent models with 3×10^5 and 0.79×10^5 backpropagation steps, respectively.

We exclusively evaluated GaitDynamics' force estimation performance on gait data of the test set. First, we excluded trials if their names explicitly contain non-gait terms such as 'jumping', 'squat', 'sit' and so on. Out of the 15 studies, 14 in the AddBiomechanics dataset were retained, with one study excluded due to the absence of gait trials. Second, we segmented the continuous gait data into individual gait cycles using a two-step process: (1) identifying tentative left foot gait cycles (from heel strike to heel strike) using vertical force with a threshold of 10% of body weight and (2) excluding gait cycles whose lengths are not within 0.2–2 s or whose stance phase lengths are not within 0.1–1.5 s. We used these conservative thresholds for gait cycle lengths to minimize the probability of excluding valid walking or running gait data. Still, we found that one of the test-set studies has a post-stroke individual whose gait cycles were substantially longer than these thresholds. Thus, for this particular study, we adjusted the length thresholds for gait cycles and stance phases to 0.1–3 s and 0.2–4 s, respectively. The peak vertical force was computed as the maximum vertical force of a gait cycle. We calculated the mean absolute error by averaging all the gait cycles within each study. We then computed the mean and one standard deviation of 14 studies' errors, ensuring that studies of different sizes contributed equally to the metrics.

Downstream task 2: predicting knee adduction moment during trunk sway gait without experiments. Only one training set study includes trunk sway gait, where participants walked with increased bilateral trunk sway on a treadmill⁴⁸. We performed testing on one test-set study⁴⁰ that was not used for model training. The study involves ten participants performing normal walking and walking with increased ipsilateral trunk sway, leaning towards the left foot. Thus, we performed the analyses on each participant's left knee. For all the normal walking trials, we created synthetic trunk kinematics ($\hat{q}_{\text{trunk}}$) by amplifying the magnitude of medial–lateral trunk angles by 2.0×, 2.5× and 3.0×, while retaining the magnitudes of the anterior–posterior trunk angles and trunk rotation angles (Fig. 5). Then, we used synthetic trunk kinematics as inputs to generate unknown kinematics and forces, following DDIM sampling and inpainting described in Downstream task 1. To ensure that the model generated walking gaits, we used guided diffusion, a method that uses data to regularize the generation process of diffusion models^{41–43}. Specifically, we designed a loss function g to regularize large discrepancies between the synthetic gait window ($\hat{x}_0$) and the denoised window ($\hat{x}_0$) in each iteration as follows:

$$g(\hat{x}_0, \hat{x}_0) = \text{SUM}(\text{ReLU}(|x_0^{\nu,q} - \hat{x}_0^{\nu,q}| - 10\% \times \text{RoM}_i)) \quad (13)$$

where ReLU is a rectified linear unit function, upper script ν, q denotes each degree of freedom of ν and q , and RoM_i is the range of motion for

the corresponding degree of freedom. As such, we did not regularize any discrepancies in joint angular velocities or forces, or small discrepancies in pelvis linear velocities or joint angles that were within 10% of the range of motion. During each inpainting iteration, the loss function g was back propagated to the noisy window five times, iteratively adjusting the window using gradient descent optimization with a learning rate of 0.02 as follows:

$$\begin{aligned}x_{k,n+1} &= x_{k,n} - 0.02 \times \nabla g(\hat{x}_{0,n}, \hat{x}_0) \\ \hat{x}_{0,n} &= \epsilon_\theta(x_{k,n}, k) \\ \text{for } n &= 1, 2, 3, 4, 5\end{aligned}\quad (14)$$

The resulting $x_{k,n+1}$ as well as k were then used as inputs for subsequent denoising (equation 8), diffusion (equation 10) and known portion replacement (equation 12). For both experimental measurements and GaitDynamics generations, we calculated knee adduction moments via the cross-product approach⁸¹ as follows:

$$\text{Knee adduction moment} = \frac{r_y \times f_z - r_z \times f_y}{\text{Body weight} \times \text{Body height}} \quad (15)$$

where r denotes the lever arm vector from the knee joint centre to the centre of pressure, and y and z are aligned with the vertical and medial–lateral directions, respectively.

We evaluated the accuracy of model predictions by comparing the measured and predicted reductions in the first and second peaks of the knee adduction moment. Specifically, we resampled the stance phase of each gait cycle to a range of 0–100%. We then calculated the mean and standard deviation of the knee adduction moment curves across all participants for experimental normal walking, experimental large trunk sway and each synthetic large trunk sway. The maximum value within 0–50% of the mean knee adduction moment curve was defined as the first peak, while the maximum within 50–100% was defined as the second peak. We computed the moment reductions as the knee adduction moment of normal walking minus that of experimental or synthetic large trunk sway.

Downstream task 3: gait generation at various running speeds.

Three training set studies included running trials with speeds ranging from 2.3 m s^{−1} to 4.7 m s^{−1} (refs. 49,59,61). We performed testing on one test-set running study⁴⁴ that was not used for model training. The study involves ten participants performing normal running on a treadmill at four different speeds: 2.0, 3.0, 4.0 and 5.0 m s^{−1}. Following the original study, we performed the analyses on each participant's right-foot gait cycles, which were segmented using vertical force with a threshold of 10% of BW. For each participant's experimental 4.0 m s^{−1} running data, we scaled the anterior–posterior component of pelvis linear velocities by factors from 75% to 125% in 10% increments, thus creating synthetic running speeds from 3.0 m s^{−1} to 5.0 m s^{−1} in 0.4 m s^{−1} increments. The vertical and medial–lateral components of pelvis linear velocities were retained from the experimental 4.0 m s^{−1} running data. We applied the method used in Downstream task 2 (Fig. 5) to generate full-body kinematics that were coherent to the synthetic running speeds. The known parameters of this task were synthetic pelvis linear velocities, instead of synthetic trunk kinematics that were used in Task 2. To generate a wide range of running speeds, the loss function g only regularizes discrepancies in pelvis linear velocities v or joint angles q when they exceed 30% of the range of motion.

We compared the performance of GaitDynamics in kinematics generation against a state-of-the-art motion generation model, OmniControl, which uses partial-body kinematics and text as inputs⁴⁵. We implemented OmniControl by directly using its open-source code and model. The kinematic inputs were pelvis (root of the skeletal model) positions computed from running speeds of 3.0–5.0 m s^{−1} in 0.4 m s^{−1}

increments. The text input was “a person runs continuously in a straight line”, consistent for all the running speeds. We did not specify the numerical value of speed in the text input since they led to more unsuccessful generations in pilot testing. We repeated the generation 10 times with different random seeds. We segmented the gait cycles of the right foot using its anterior–posterior pelvis linear velocity with a threshold of 0.1 m s^{−1}. We manually reviewed generations with no gait cycle detected and removed the generation if we confirmed that the generation was unsuccessful (that is, not human running gait). For the successful generations, we first computed the average stride length and peak knee flexion angle across gait cycles within each generation, and then computed the mean and one standard deviation across generations of each speed.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

This work utilized the AddBiomechanics dataset that consists of data from 15 previous studies (https://addbiomechanics.org/download_data.html). Source data are provided with this paper.

Code availability

The trained models and source codes are available on GitHub at <https://github.com/stanfordnbl/GaitDynamics> (ref. 82). We also provide a Hugging Face demo for users to upload files with full- or partial-body kinematics, predict ground reaction forces and missing kinematics using GaitDynamics, and download the results (<https://huggingface.co/spaces/alanttan/GaitDynamics>). Note that we used the an OpenSim Rajagopal Model without Arms (https://simtk.org/projects/full_body)⁶⁸, and a different skeletal model may lead to failure due to mismatching joint names.

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Author contributions

All authors contributed to the conception of the work. T.T., T.V.W. and K.F.W. worked on data analysis. T.T. drafted the manuscript. All authors critically revised the manuscript and approved the final version.

Competing interests

The authors declare no competing interests.

Additional information

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